

MECHANICAL PROPERTIES OF FLUIDS

What is a Fluid?

A fluid is a substance which can flow and take the shape of its container.
Fluids include liquids and gases.



Takes the shape of container

Liquids

Have fixed volume but no fixed shape.

Gases

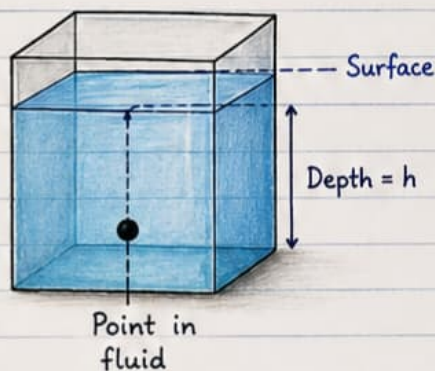
Have neither fixed shape nor fixed volume.

1. Pressure in Fluids

- Fluids exert **pressure** on the walls of the container and on the objects immersed in them.
- The pressure at a point in a fluid at rest depends on the **depth** of the point below the surface.

$$\text{Pressure at depth } h : P = \rho gh$$

where, ρ = density of fluid
 g = acceleration due to gravity
 h = depth below the surface

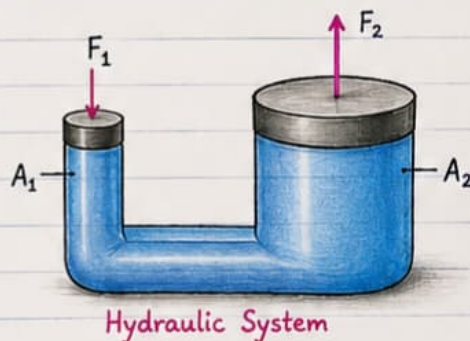


2. Pascal's Law

Pressure applied at any point in an enclosed fluid is transmitted **equally and undiminished in all directions** throughout the fluid.

Used in hydraulic lift, brakes, presses etc.

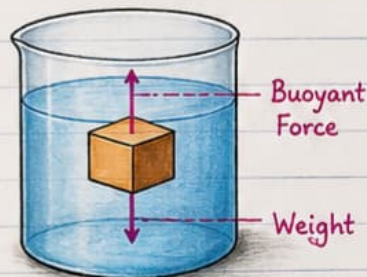
$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$



3. Buoyancy and Archimedes' Principle

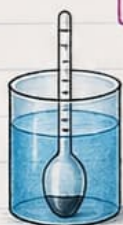
- When a body is partially or fully immersed in a fluid, it experiences an upward thrust called **buoyant force**.

Archimedes' Principle : A body wholly or partially immersed in a fluid experiences an upward buoyant force equal to the weight of the fluid displaced by it.



$$F_B = \text{Weight of fluid displaced} \\ = \rho_{\text{fluid}} \times g \times V_{\text{displaced}}$$

ρ_{fluid} = density of fluid
 $V_{\text{displaced}}$ = volume displaced



Examples of Buoyancy

- Ships floating on water
- Hot air balloons rising in air
- Hydrometers to measure density of liquids



4. Density and Specific Gravity

- Density (ρ) of a fluid is its mass per unit volume.

$$\rho = \frac{\text{mass}}{\text{volume}} = \frac{m}{V}$$

- SI unit : kg m^{-3}
- CGS unit : g cm^{-3}

Specific Gravity (S.G.):

It is the ratio of density of a substance to the density of water at 4°C .

$$\text{S.G.} = \frac{\rho_{\text{substance}}}{\rho_{\text{water}}}$$

S.G. is a pure number (no unit).

EXAMPLE

Find S.G. of a liquid whose density is 800 kg m^{-3} .

$$\rho_{\text{water}} = 1000 \text{ kg m}^{-3}$$

$$\text{S.G.} = \frac{800}{1000} = 0.8$$



Water ($\rho = 1000 \text{ kg/m}^3$) Oil ($\rho = 800 \text{ kg/m}^3$)

5. Equation of Continuity

- It is based on the principle of conservation of mass.
- For steady flow of an incompressible fluid through a tube of non-uniform cross-section, the mass flowing through any cross-section per second is constant.

Mathematically,

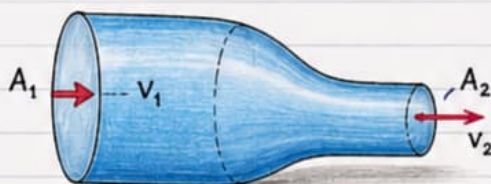
$$A_1 v_1 \rho = A_2 v_2 \rho$$

$$A_1 v_1 = A_2 v_2$$

where,

A = cross-sectional area

v = speed of fluid



EXAMPLE

Water flows through a pipe of area 10 cm^2 with speed 2 m s^{-1} . It enters into a narrow section of area 5 cm^2 . Find the speed in the narrow section.

$$A_1 v_1 = A_2 v_2$$

$$(10)(2) = (5) v_2$$

$$v_2 = 4 \text{ m s}^{-1}$$



Key Point : If area decreases, speed increases and vice versa.

6. Bernoulli's Theorem

- It states that for an ideal fluid (incompressible, non-viscous) in steady flow along a streamline, the sum of pressure energy, kinetic energy and potential energy per unit volume remains constant.

$$P + \frac{1}{2} \rho v^2 + \rho g h = \text{constant}$$

$\downarrow$ Pressure energy
 $\downarrow$ Kinetic energy
 $\downarrow$ Potential energy

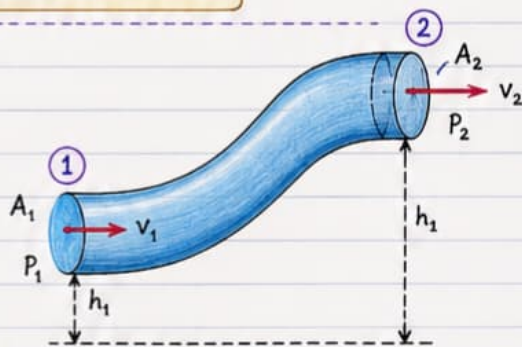
P = Pressure

ρ = Density

v = Speed of fluid

g = Acceleration due to gravity

h = Height

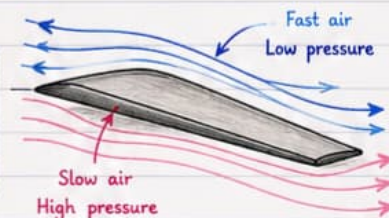


Implications :

- If speed increases, pressure decreases.
- If height increases, pressure decreases.

Real-life Applications

- Airplanes generate lift due to difference in pressure above and below the wings.
- Spray pumps, carburetors, pitot tubes work on Bernoulli's principle.



EXAMPLE

In a horizontal pipe, speed of water at a section is 3 m s^{-1} and pressure is $1.5 \times 10^5 \text{ Pa}$. If speed increases to 6 m s^{-1} , what will be the pressure at that section?

Using Bernoulli ($h_1 = h_2$):

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

$$1.5 \times 10^5 + \frac{1}{2} (1000) (3)^2 = P_2 + \frac{1}{2} (1000) (6)^2$$

$$1.5 \times 10^5 + 4500 = P_2 + 18000$$

$$P_2 = 1.365 \times 10^5 \text{ Pa}$$

★ NCERT Important Line

Fluids exert pressure on the walls of the container and on the objects immersed in them.

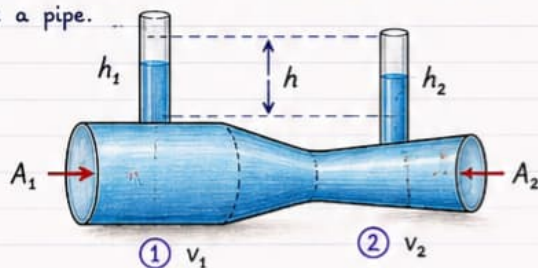
Pressure at a point in a fluid acts equally in all directions.

7. Application of Bernoulli's Theorem

Bernoulli's theorem has several practical applications.

(a) Venturimeter (To measure speed of a liquid)

- A device used to measure the speed of liquid in a pipe.



From the above, speed of liquid at narrow section,

$$v_2 = \sqrt{\frac{2gh}{1 - (A_2/A_1)^2}}$$

Using Bernoulli's theorem,

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

Continuity equation,

$$A_1 v_1 = A_2 v_2$$

EXAMPLE

In a venturimeter, $A_1 = 10 \text{ cm}^2$, $A_2 = 2 \text{ cm}^2$.

Difference in mercury level $h = 20 \text{ cm}$.

Find speed of water. ($\rho_{\text{water}} = 1000 \text{ kg m}^{-3}$)

$$v_2 = 4.4 \text{ m s}^{-1}$$

(b) Spray Gun (Atomiser)

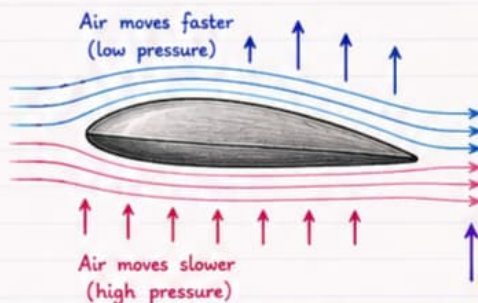
- Liquid is forced with high speed through a small hole. Due to low pressure, liquid from the bottle is sucked up and breaks into tiny droplets.



Fast moving air makes pressure low and liquid is sucked up.

(c) Lift on Aeroplane Wings

- The shape of the wing is such that air moves faster over the upper surface than below it.
 - By Bernoulli's theorem, pressure above the wing is less than below.
- This pressure difference produces an upward force called lift.



Low pressure above and high pressure below creates lift.

Resultant lift (upward force)

8. Viscosity

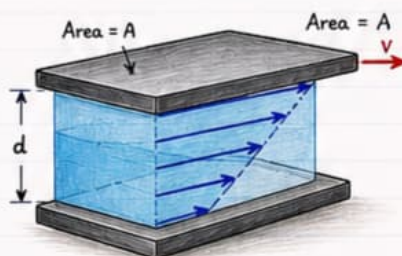
- It is the property of a fluid due to which it opposes the relative motion between its layers.
- Greater the viscosity, thicker is the fluid.

Coefficient of viscosity (η):

The tangential force (F) required to move one layer of area A with speed v over another layer at distance d is

$$F = \eta A \left(\frac{v}{d} \right) \quad (\text{SI unit : Pa s})$$

- Liquids have viscosity, Gases have very low viscosity.



EXAMPLE

A layer of oil of area 2 m^2 is moving with a speed 0.2 m s^{-1} over another layer at distance 1 mm . If tangential force required is 0.4 N , find coefficient of viscosity.

$$\eta = 2 \text{ Pa s}$$

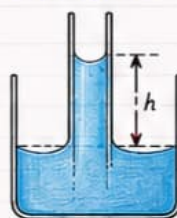
9. Surface Tension

- It is the property of liquid surface due to which it behaves like a stretched elastic sheet.
- It is due to cohesive forces between liquid molecules.
- SI unit : N m^{-1}

Effects of surface tension :

- Water drops are spherical.
- Small insects can float on water surface.
- Soap bubbles are formed.
- Rising of water in thin capillary tubes.

Capillary rise



Height of capillary rise is given by

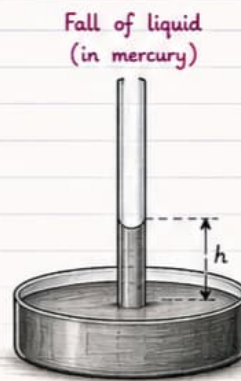
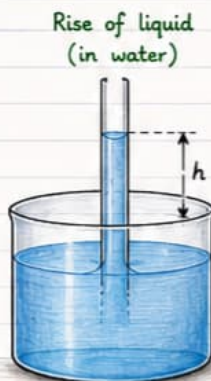
$$h = \frac{2T \cos \theta}{r \rho g}$$

where, T = surface tension
 r = radius of tube
 θ = angle of contact
 ρ = density of liquid
 g = acceleration due to gravity



10. Capillarity

- Capillarity is the rise or fall of liquid in a thin tube (capillary) due to surface tension.
- If **adhesive force** > **cohesive force**, liquid rises in the tube.
- If **cohesive force** > **adhesive force**, liquid falls in the tube.



Height of capillary rise or fall is given by

$$h = \frac{2T \cos \theta}{r \rho g}$$

where,

T = surface tension
 r = radius of capillary tube
 θ = angle of contact
 ρ = density of liquid
 g = acceleration due to gravity

Examples of Capillarity

- ✓ Water rising in thin glass tubes.
- ✓ Kerosene rising in the wick of a lamp.
- ✓ Sap rising in xylem vessels of plants.



Key Point

Smaller the radius (r) of the capillary tube, greater is the capillary rise (h).

11. Streamline Flow and Turbulent Flow

(a) Streamline Flow :

- When a fluid flows in smooth layers so that the layers do not mix with each other, the flow is called **streamline (laminar) flow**.
- It occurs at low speeds and in viscous fluids.



(b) Turbulent Flow :

- When the fluid particles move in a zig-zag manner and different layers mix with each other, the flow is called **turbulent flow**.
- It occurs at high speeds and in less viscous fluids.



Transition :

The change from streamline flow to turbulent flow occurs when the speed exceeds a certain value called critical speed.

Factors affecting nature of flow

- Speed of fluid
- Viscosity of fluid
- Radius of the tube

Reynolds Number (R_e) :

To predict the nature of flow, a dimensionless quantity called Reynolds Number is used.

$$R_e = \frac{\rho v d}{\eta}$$

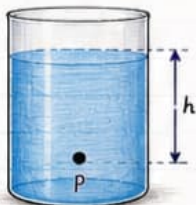
ρ = density of fluid
 v = speed of fluid
 d = diameter of tube
 η = coefficient of viscosity

Condition for flow	Reynolds Number (R_e)	Nature of flow
	$R_e < 2000$	Streamline (laminar) flow
	$2000 < R_e < 3000$	Transitional flow
	$R_e > 3000$	Turbulent flow

Applications

- Streamline flow : Oil in engine pipes, blood in wide arteries.
- Turbulent flow : Smoke from chimney, water in waterfalls.

12. Pressure Due to Column of Liquid



Consider a liquid column of height h and base area A .

$$\text{Pressure at depth } h, \\ P = \rho g h$$

It depends on :

- ✓ Density of liquid (ρ)
- ✓ Depth (h)
- ✓ Independent of shape and area of container

13. Bulk Modulus (K)

- It is the property of a fluid due to which it opposes the change in its volume when pressure is applied.
- Less compressible fluids have high bulk modulus.

$$\text{Bulk modulus } (K) = \frac{\text{Increase in pressure}}{\text{Decrease in volume / Original volume}}$$

$$\text{OR } K = -V \frac{dP}{dV}$$

SI unit : Pa (Pascal)

SUMMARY

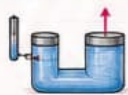
①

Fluids exert pressure in all directions and at depth h ,
 $P = \rho g h$



②

Pascal's law : Pressure applied to enclosed fluid is transmitted equally in all directions.



③

Buoyancy : Upthrust on a body equals weight of displaced fluid.



④

Bernoulli's theorem :

$$P + \frac{1}{2} \rho v^2 + \rho g h = \text{constant}$$



⑤

Surface tension and viscosity are important properties of fluids.



Remember !

- Understand concepts
- Draw neat diagrams
- Practice numericals
- Revise regularly

